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A Hybrid Power System Laboratory

Testing electric and hybrid propulsion.

INTERNATIONAL SHIPPING CONTRIBUTES to a major portion of global emissions. This is expected to grow at a faster pace due to economic and population growth. According to the Third Greenhouse Gas (GHG) Study 2014 of the International Maritime Organization (IMO), the total shipping emissions during 2007–2012 were 3.1% of global CO₂ emissions, 2.8% of global CO₂-equivalent (CO₂e) emissions, 13% of global sulfur oxide (SO_x) emissions, and 15% of global nitrogen oxide (NO_x) emissions. Further details are given in Table 1.

The IMO is implementing strategies to reduce the emissions from international shipping by establishing

standards, imposing restrictions, and providing incentives. The initial strategy envisages a reduction in total GHG emissions from international shipping by at least 50% by 2050 compared to 2008, consistent with the Paris Agreement. IMO's current global standard for SO_x emission is 3.5% mass by mass (m/m). IMO has set a new limit of 0.5% m/m by 2020 per the International Convention for the Prevention of Pollution by Ships (MARPOL 73/78) Annex VI. Similarly, the emission limit for NO_x from engines is dependent on the type of the engine and the nominal engine speed. The global limit has been set to 14.4 g/kilowatt hour (kWh) for engines with speeds less than 130 r/min and 7.7 g/kWh for engines with speed greater than 2,000 r/min as shown in Figure 1(b).

To comply with the standards and emission reduction goals set by IMO, the maritime industry is looking for

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TABLE 1. Total shipping emissions (average value per annum during 2007–2012).

Parameter	CO ₂ Emissions	CO ₂ e Emissions	SO _x Emissions	NO _x Emissions
Quantity (million tons)	1,015	1,036	11.3	20.9
% of global emissions	3.1	2.8	13	15

reduced- and zero-emission technologies and approaches. The transition from traditional fossil fuel engines to zero-emission technologies is challenging in a competitive market, where manufacturers are struggling to increase revenue and reduce production cost. The incentives to reduce emissions and improve the efficiency of equipment are motivating ship manufacturers to adopt zero-emission technologies. Fuel cells and energy storages—for example, batteries and supercapacitors—reduce GHG emissions and are key enablers for the realization of low-emission operation of marine vessels.

Both academic institutions and the maritime industries are working hand in hand to innovate solutions that are cleaner, more efficient, and more effective. Recently, China launched a fully electric cargo ship driven by the combination of supercapacitors and lithium batteries with the total energy capacity of 2.4 MW/h. Small-route ferries are being converted from fossil fuel-driven engines to the hybrid type or fully electric vessels, such as the car and

passenger ferry MF Ampere in Norway and ferries operated by Sweden's HH Ferries Group. Hurtigruten in Norway is building cruise ships driven by the hybrid power plant using a battery system that reduces the fuel consumption and CO₂ emission by 20%. Kongsberg Group, one of the advanced players in the maritime industry, is working side by side with academic institutions to innovate these new ideas. Kongsberg Group has recently been involved, along with Yara, in the development of the zero-emission container feeder *Yara Birkeland*, which will be fully autonomous, in addition to fully battery-powered, reducing its emissions by the equivalent of 40,000 truck journeys per year.

Research infrastructure and facilities are a critical requirement to test new solutions and technologies that enable reduced- and zero-emission marine transportation. The Hybrid Power Systems (HPS) laboratory at the Department of Marine Technology of the Norwegian University of Science and Technology (NTNU) is an infrastructure for research and testing of marine electric power and propulsion based on a dc electric power system. In this lab, the electric power required by the propulsion loads is supplied by a hybrid power system that includes generator sets along with fuel cells, batteries, and supercapacitors. The propulsion loads are connected to the main dc power system through converter interfaces. This enables testing of the paradigm shift of ship propulsion from traditional mechanical propulsion to electric propulsion as well as the change of onboard energy scenarios from traditional diesel engines to reduced- and zero-emission energy solutions through hybrid power systems.

HPS Laboratory Architecture

The layout diagram of the HPS lab at NTNU is shown in Figure 2. It has two variable-speed, diesel engine generator sets with different capacities connected to a dc distribution system through rectifiers. The dc grid is also supplied by a fuel cell stack. Furthermore, the dc grid is connected to energy storage devices (ESDs), i.e., battery and supercapacitor banks, through dc-dc converters. Propulsion loads of the vessel are emulated via two electric consumers, each comprising a variable-speed electric motor driving an eddy current brake. These electric consumers can be dynamically loaded through a load controller where it is possible to select different set points, both for the speed and torque. The set points can be fixed values or functions like pulse, ramp, sinusoidal, and so on.

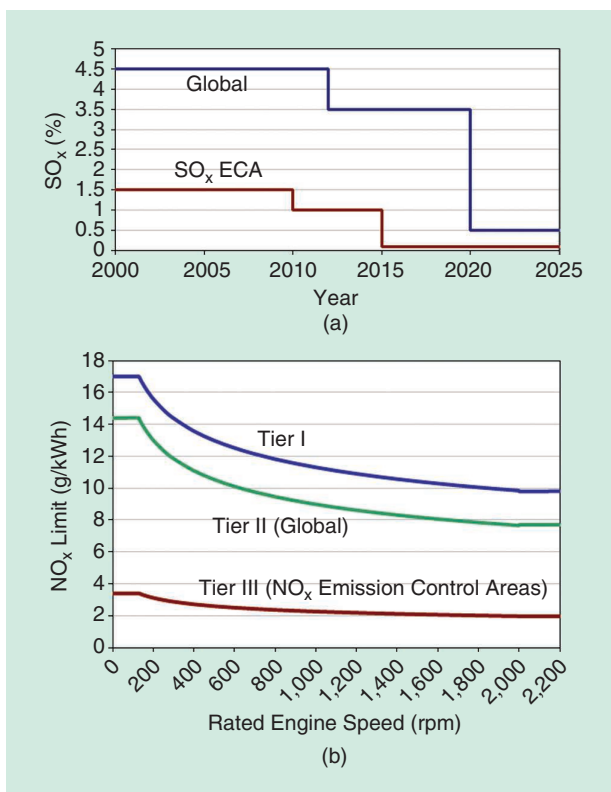


Figure 1. The MARPOL Annex VI emission limits for global and Emission Controlled Areas (ECAs) perspective (a) SO_x and (b) NO_x. (Source: DieselNet, www.dieselnet.com; used with permission.)

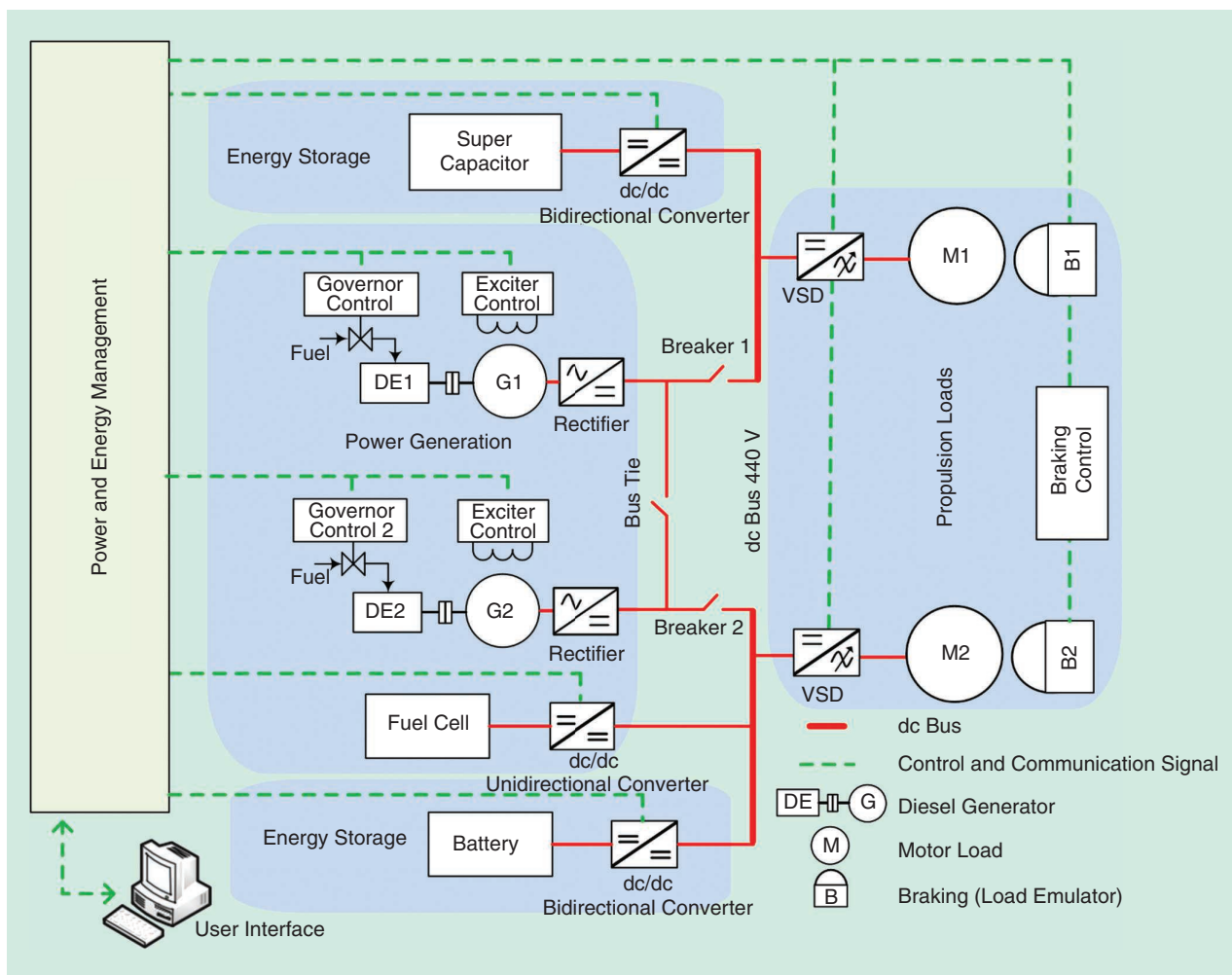


Figure 2. A single-line diagram of the hybrid power lab based on onboard dc power system.

Traditionally, the ac power grid has dominated marine power plants due to the complexity in the development of safety components and systems in the dc power grid. However, dc power grids are becoming popular due to the fast advancement in the power electronic technologies and improved safety systems. The general benefits and the driving factors for adopting dc power grids include improved prime mover's efficiency; reduction in average fuel consumption; low overall transmission losses; faster and simpler parallel connection of generators; and simpler implementation of ESDs.

Fuel cells, combustion engines, batteries, and supercapacitors are the most common energy carrier solutions that are in discussion for the hybrid power system. Selection of an energy carrier solution for a marine vessel depends on the mission of the vessel as well as the characteristics of the energy carriers. Energy density and power density are a couple of the major characteristics that are necessary to be considered. A simplified Ragone plot in Figure 3 shows the range of the power density versus energy density in a log-log scale for different energy

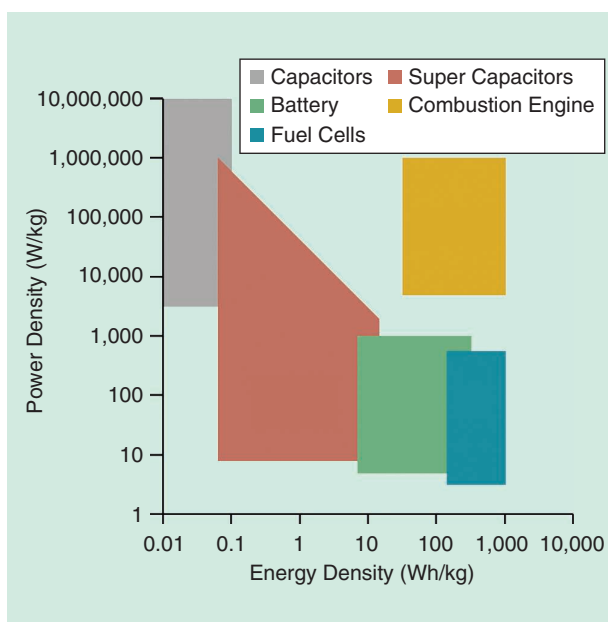


Figure 3. A simplified Ragone plot showing a range of power density versus energy densities for different energy carriers, such as sources and storages.

TABLE 2. The specifications of components.

Equipment	Specifications
Big engine (Perkins 2506C-E15TAG1)	400 kVA, 900–1,800 r/min
Small engine (Perkins 1306C)	230 kVA, 900–1,800 r/min
Engine controller (governor)	CAT C15
Fuel cell stack	60 kW
Battery bank	Li-Ion, 159 Ah/45 kWh, 346 V
Capacitor bank	1 MW/s, (3 kF × 200 pcs)
Electric motor loads—induction motor	160 kW, 1,015 nm at 1,500 r/min
dc bus voltage	560 V dc
Electric braking (eddy current)	1,000 Nm max braking
Data acquisition system	10 Hz, Labview NI cRIO
NO _x emission sensor	Horiba Mexa-720

solutions. It demonstrates that the capacitors have the highest power density but the least energy density. At the same time, fuel cells have the highest energy density but the least power density. Batteries lie somewhere in between them. This signifies that capacitors, in general, can output very high power per mass, but their energy content per mass is much less. Fuel cells can output the least power per mass; however, their energy content per mass is the highest. On the other hand, a battery can deliver higher power per mass than the fuel cell and contains higher energy per mass than the capacitor. The range of energy and power density for energy storage solutions is one of the criteria used for deciding their functionality.

ESDs in a marine vessel enhance the optimal operation of conventional engines and hence reduce emission and fuel consumption. The major roles or functionalities of ESDs in a marine power plant are spinning reserve, peak shaving, enhanced dynamic performance, enhanced ride through, strategic loading, and zero-emission.

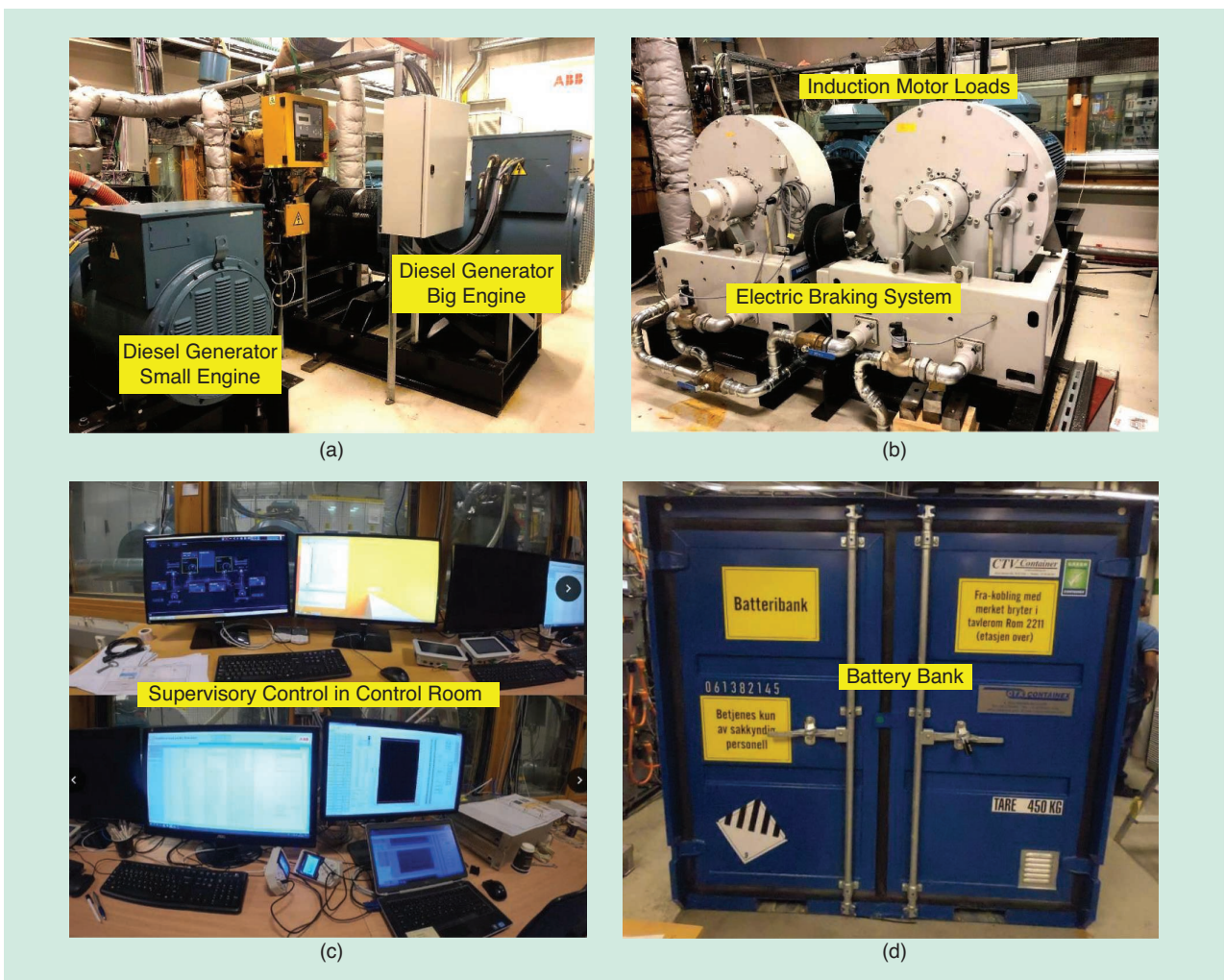


Figure 4. (a) A variable-speed diesel engine and generator. (b) The eddy current brakes along with the driving variable-speed motors. (c) The HPS lab control room. (d) The battery bank.

The specifications of some major equipment and energy carriers in the hybrid power lab are listed in Table 2. The infrastructures of the HPS laboratory are distributed between a lab machinery room, dedicated control room, and battery bank compartment, along with an external compartment for the fuel cell. The power generated by two 30-kW modules of proton exchange membrane fuel cells is connected to the hybrid lab, which is considered a most suitable type of fuel cell for marine transportation due to low operating temperature, high power density, and modularity (or scalability). Figure 4 includes photos from the HPS lab machinery room. Figure 4(a) displays a variable-speed engine along with the generator. Similarly, 4(b) illustrates the eddy current brakes along with the driving variable-speed motors for emulating the marine loads. The hybrid power lab is fully equipped to be monitored and operated from the control room. Several sensors in the equipment harvest the real-time data that are then made available for research and studies in the control and monitoring of computers.

Research Infrastructure in the HPS Laboratory

The main objective of the HPS lab is to enhance research and research-based education in the field of marine power systems and marine electrification. The lab is used by researchers to verify theoretical methods and controllers. In addition, it is used in graduate courses related to marine electric power and propulsion. The HPS lab offers several types of test experiments ranging from new control strategies to controlling the entire power plant. The load emulator generates load fluctuations similar to the real propulsion load by applying real-case scenarios of the ship load profile. Control algorithms are developed for the hybrid energy system to reduce fuel and emissions through a so-called energy- and emission-management system. By applying a control algorithm via the user interface to the central controller or supervisory controller of the power

management system, experiments such as the following can be conducted:

- ▲ **Load sharing:** This involves sharing the total load power between two generators connected in parallel. Load sharing may be implemented based on two control approaches, such as isochronous and droop approach. Also, load sharing between generators and battery can be done. This will further be discussed in the section “Load-Sharing Scenario.”
- ▲ **Peak shaving/power smoothing:** Generator sets are connected to the load with the battery and the supercapacitor for peak shaving/power smoothing. The power smoothing by the battery and supercapacitor provides the instantaneous power for the peak power during load fluctuation, and the generator will be unable to serve due to the delay from the mechanical system dynamic. Thus, the diesel generators (DGs) generate the constant average power, and the rest is supplied by battery and supercapacitor. Due to the high power density characteristic of the supercapacitor, it can provide instant high power with a short time to the load for high power smoothing, while the battery provides the rest. It is also used

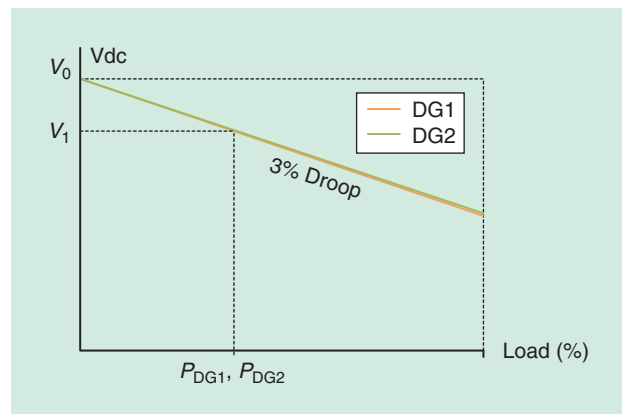


Figure 5. A graph demonstrating equal droop power sharing with two generators.

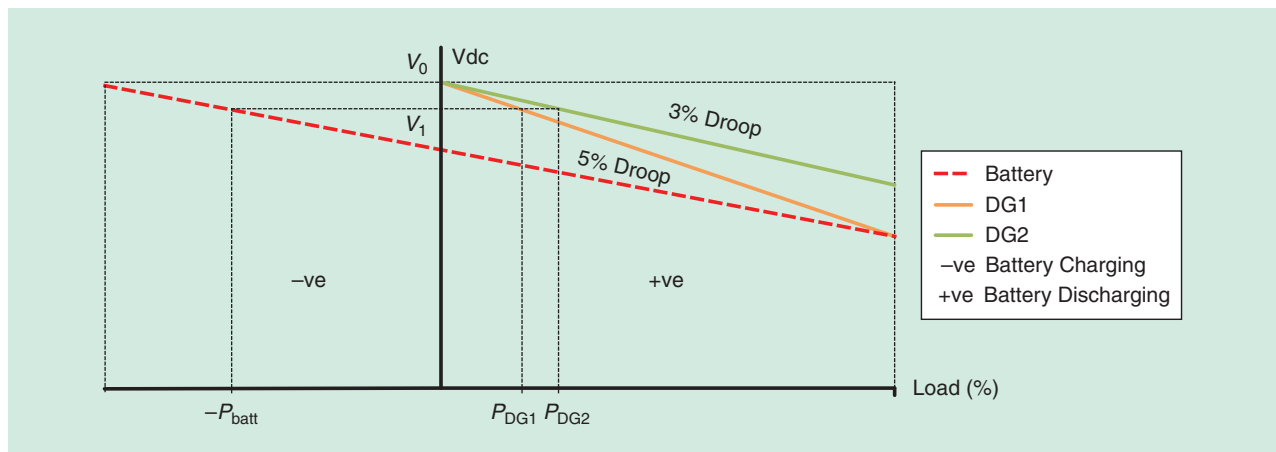


Figure 6. A graph indicating unequal droop power sharing with two DGs and a battery.

to compensate for high-frequency load fluctuations. However, to provide the power smoothing larger than one min, other devices with higher-energy content and longer duration response, such as the battery, can be activated.

- Strategic loading:** This tests the new control strategy or algorithm to optimize the operating point of the DG with and without the estimated standard deviation. For example, to avoid being operated in a low-load factor, DGs could charge the battery (in addition to serving the load) so as to reach optimal efficiency point based on the specific fuel consumption curve (SFC) of the engine. This study also demonstrates the charging and discharging role of the onboard batteries while satisfying the operating point of the DG.

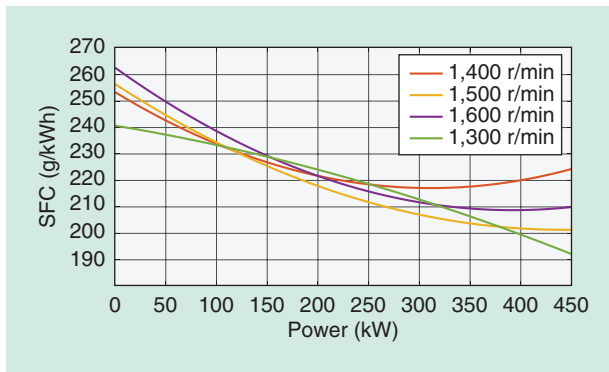


Figure 7. The SFC curve fitting of different speeds, plotted based on the real test case of DG-Perkins2506C.

- Zero-Emission propulsion:** This experiment operates by running the load by only using the ESD (battery) or fuel cell to demonstrate zero-emission operation where the diesel generators are not dispatched.

Load-Sharing Scenario

In a multisource system, load sharing among parallel generators/sources based on the droop concept mode has been well established. Based on the hierarchical control scheme in power systems, load sharing can be done at a primary level using local controllers. For example, in the ac distribution system, active and reactive power sharing are done by regulating the frequency and voltage, respectively, performed by the local control. Nevertheless, due to the absence of frequency in dc distribution as well as reactive power, voltage is the only variable that can be controlled to realize the load sharing based on droop control. Therefore, frequency does not bring any effect to the power sharing in the dc system. Voltage control in dc can be realized by using an automatic voltage regulator (AVR) or an active rectifier (controlled rectifier). Voltage regulation using AVR is more cost-effective as compared to the active rectifier. However, it could not perform the decoupling control as in the active rectifier. Thus, any transient voltage that happens in ac generation side can bring effect to the dc side. In this laboratory facility, the voltage control for the dc bus is performed by the generator excitation or AVR. Figure 5 shows the equal load sharing between two generators. The droop coefficient is the variable to control the DG to have equal or unequal load sharing. Furthermore, the droop coefficient could be adjusted to have

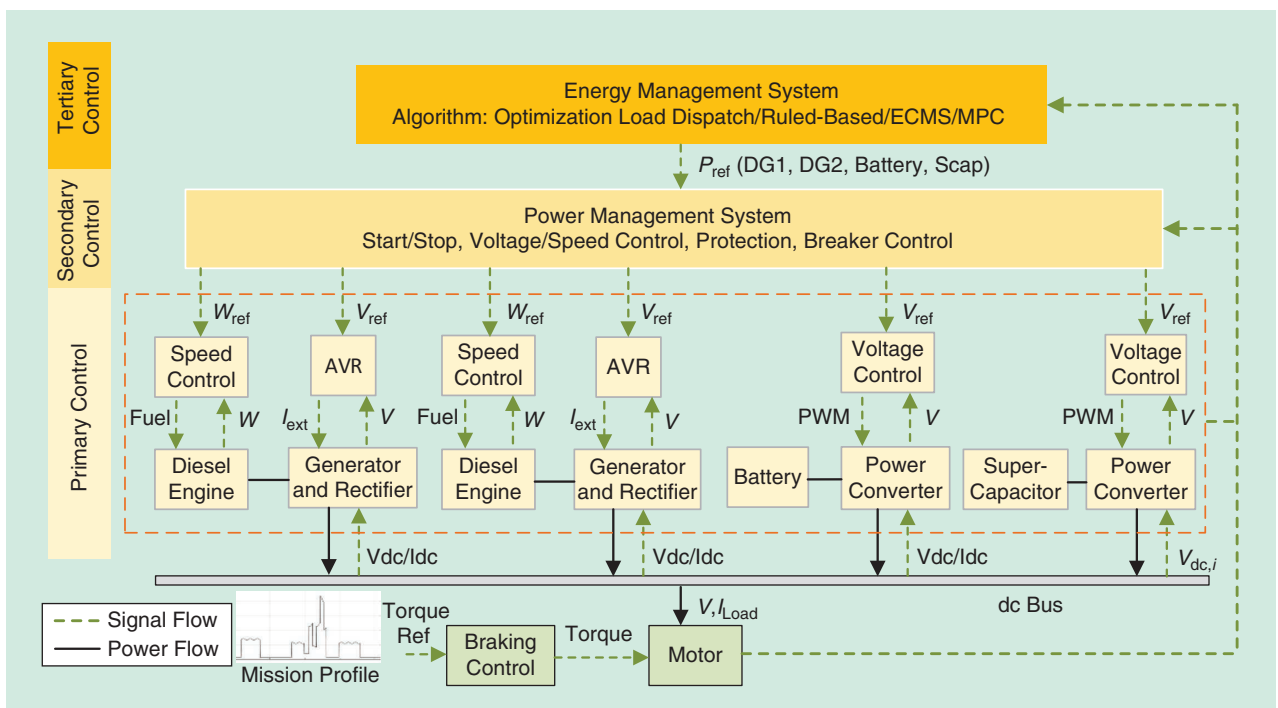


Figure 8. A simplified layout of a control system in a hybrid shipboard power system laboratory. ECMS: equivalent consumption minimization strategy.

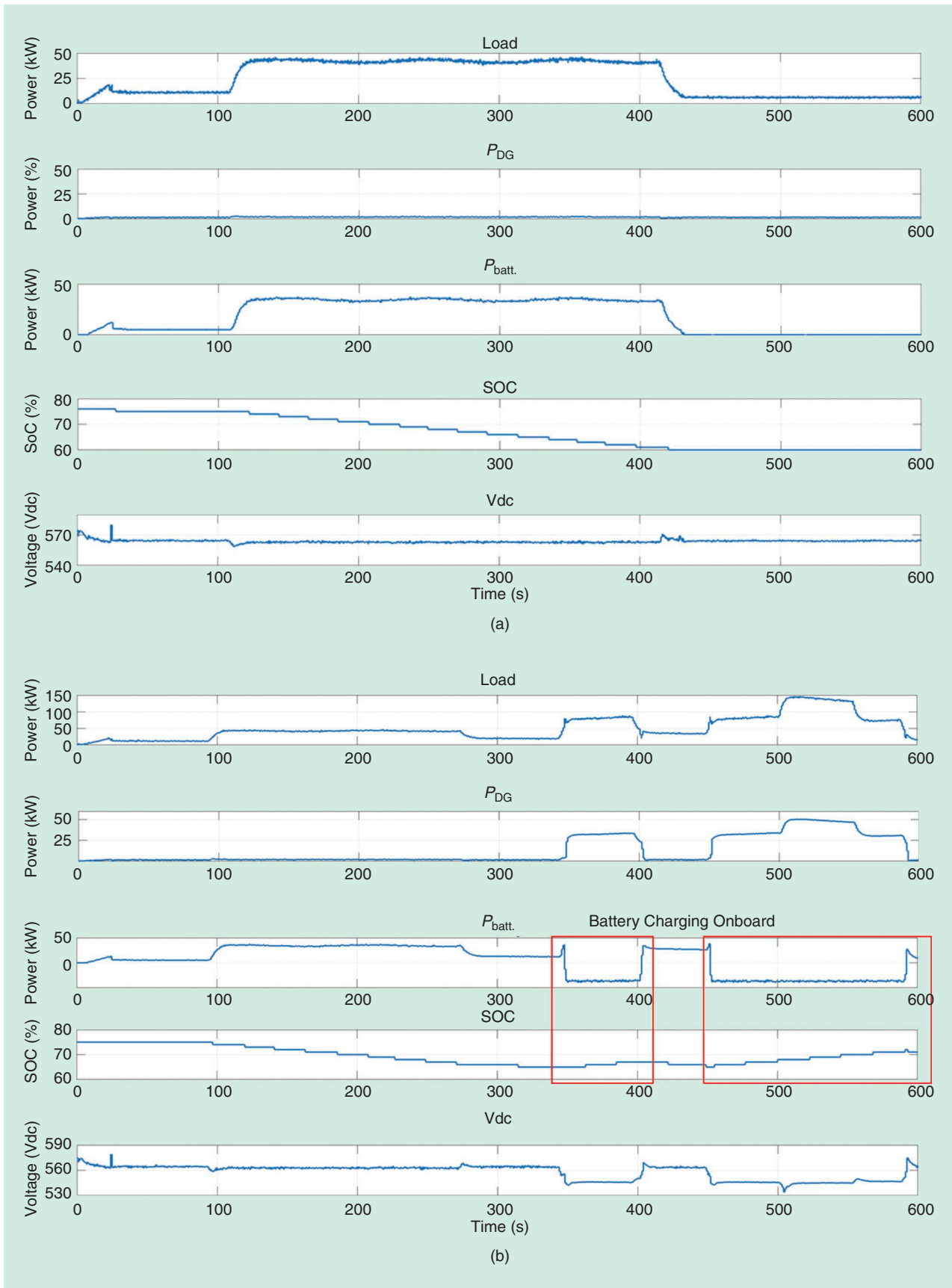


Figure 9. Experimental verification of the energy management system with real ship load-profile: (a) Zero-emission mode with the battery during berthing. (b) Hybrid propulsion mode with the diesel generator and battery (demonstrate battery charging/discharging).

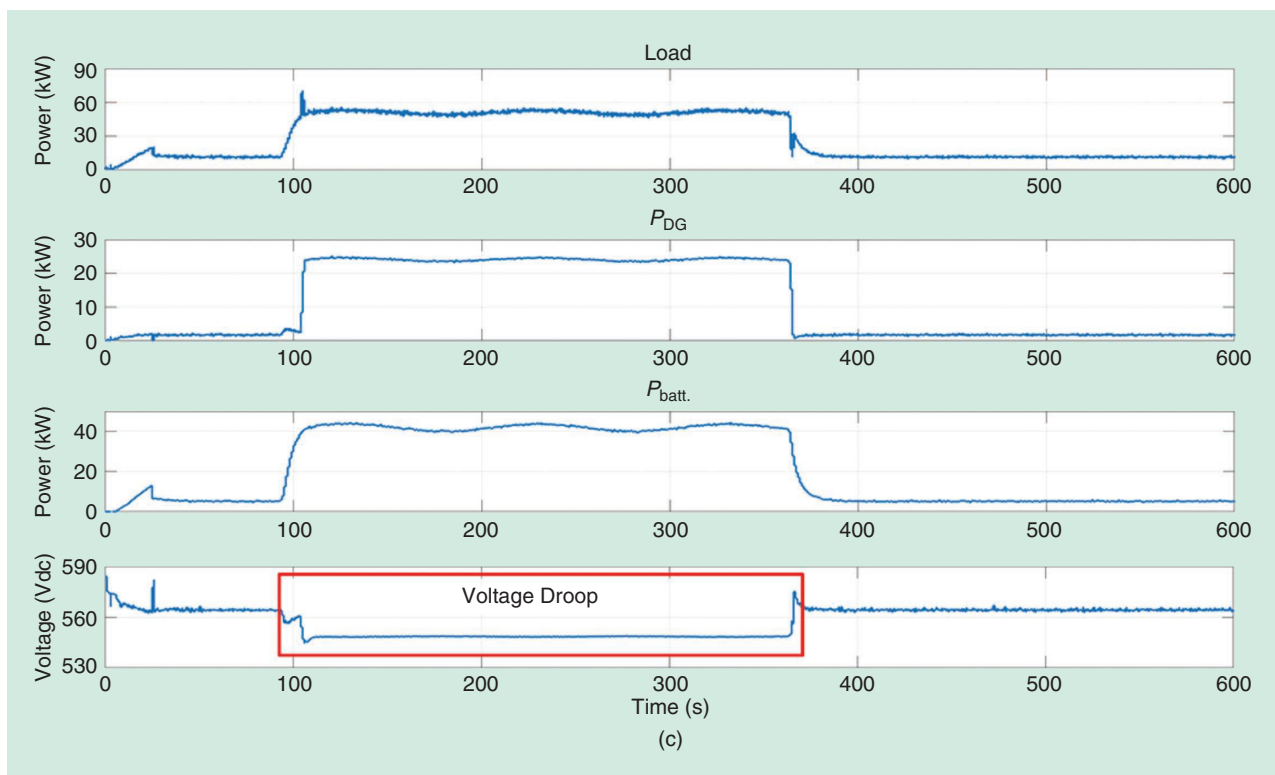


Figure 9. (Continued) (c) A demonstration of voltage droop power-sharing DG and battery. SOC: state of charge.

better fuel efficiency based on DG's SFC map. Figure 6 demonstrates the example of unequal load sharing between DGs and the battery. The left side, which is denoted as “-ve,” represents the charging region of the battery.

Fuel Consumption

Fuel consumption is estimated based on the engine's SFC map, which is provided by the engine manufacturers. Figure 7 shows the curve fitting of a Perkins 2506C with a rated capacity of 460 kW, taken from the experimental data. Unlike an ac system, the variable speed of DG in the dc system is proven to have better fuel consumption based on the SFC curve. Due to the absence of frequency in the dc system, the engine speed can be varied. Therefore, the minimum SFC can be achieved by varying the engine speed to the optimal point. Moreover, DGs are built in such a way to have high efficiency when connected to a 70–90% load factor.

Lower fuel consumption can be achieved by using strategic loading to the DG. For example, during low-load conditions, the battery can be discharged while the DG is off. When the load is high, the battery can share the load with the DG to avoid another DG to be dispatched with a low-load factor. Furthermore, during low-load conditions and when the battery has a low state of charge (SOC), DG can serve the load and charge the battery. To shift the DG's operating point to the lowest SFC, excess power from the generator can charge the battery.

Power and energy management in Figure 8 demonstrates the classification of the control strategy of the HPS lab. Due to the similarity of a shipboard power system and a stand-alone microgrid, a similar hierarchical control scheme can be implemented. The lowest control level, called *primary control*, will perform a local control, i.e. voltage deviation or load power-sharing (such as droop or isochronous mode). The next level, called *secondary control* (power management system), performs the voltage restoration introduced by primary control and performs control action for ensuring the security of the system in meeting the load demand. The highest control level, called *tertiary control*, has more responsibilities to provide the power reference based on optimal strategy or power scheduling, including the rule-based, equivalent consumption minimization strategy (ECMS), optimization algorithm and so on. At this level, the control strategy can be done by including specific optimization objectives (e.g., minimizing fuel costs or reducing emissions) and by considering several technical constraints (battery SOC limit, generator ramp-rate limit, startup shut-down limit, and so on). All individual control levels are connected to the secondary and tertiary control by the communication signal, which carries all of the measurements from the sensors.

Experimental Verification

Figure 9 shows a sample of experimental results conducted in the HPS lab for the testing of the ship power management, with a predefined rule-based strategy in

the supervisory controller. In this experiment, the real load profile based on a tugboat operation is emulated by the eddy current brake loads and is used as a case study to test the power and energy management system. Figure 9(a) demonstrates the battery used to serve the load while maintaining the dc bus voltage. Based on Figure 9(b), there is a change of power dispatch between DG and battery to serve the load. At a lower load demand, the battery will be used to serve the load. The onboard battery charging by DG is also demonstrated. At this point, the battery will be charged so that the load on the generator can be increased to prevent the generator from operating under low-load conditions. DG can be adjusted to the optimal operating point by adjusting the charging power of the battery. Figure 9(c) indicates the power sharing between DG and battery to serve the load. In this case, droop mode power sharing is demonstrated where dc voltage drops within 6% and the voltage of dc bus is maintained at 550–570 Vdc.

Conclusions

Onboard hybrid power systems with low-emission energy carriers are a promising solution for green shipping, replacing conventional diesel engine-based energy systems. However, there are several existing challenges related to these new low-emission energy solutions that need to be investigated. A full-scale infrastructure for testing and verification of such systems is one of the important stepping stones while developing a reliable system. In this article, hybrid power systems for ships and the related controllers were discussed with a focus on the HPS Laboratory at NTNU as an infrastructure for the testing and verification of shipboard dc power systems. A hierarchical control architecture for the power and energy management of the hybrid power system was also detailed. The experimental verification of the energy management system for both the zero-emission and hybrid propulsion mode along with voltage droop power-sharing using the real ship load profile was presented.

The HPS laboratory can effectively contribute to the research and development of zero-emission shipping solutions through the development of energy- and emission-management systems for hybrid power systems and technology development of shipboard microgrids. Moreover, the lab infrastructure could accommodate the development of new control strategies and perform hardware-in-the-loop testing under the operating conditions of the actual ship operation. Various real-field tests can also be conducted on the laboratory setup based on the real case mission profile, including the battery systems, supercapacitors, and fuel cells.

For Further Reading

J. M. Guerrero et al., "Shipboard microgrids: Maritime Islanded power systems technologies," in *Proc. PCIM Asia*, June 2016, pp. 28–30.

N. P. Reddy, D. Padeloup, M. K. Zadeh, and R. Skjetne, "An intelligent power and energy management system for fuel cell/battery hybrid electric vehicle using reinforcement learning," in *Proc. 2019 IEEE Transportation Electrification Conf. and Expo (ITEC)*, pp. 1–6.

D. Park and M. K. Zadeh, "Dynamic modeling and stability analysis of onboard DC power system for hybrid electric ships," in *Proc. 2019 IEEE Transportation Electrification Conf. and Expo (ITEC)*, pp. 1–6.

M. Bin Othman Marzuki, A. Anvari-Moghaddam, N. Baidura Binti Ahamad, C.-L. Su, and J. M. Guerrero, "Scheduling of power generation in hybrid shipboard microgrids with energy storage systems," in *Proc. IEEE 18th Int. Conf. Environment and Electrical Engineering and 2nd Industrial and Commercial Power Systems Europe*, 2018, pp. 1–6.

M. K. Zadeh, L.-M. Saublet, R. Gavagsaz-Ghoachani, S. Pierfederici, B. Nahid-Mobarakeh, and M. Molinas, "Energy management and stabilization of a hybrid DC microgrid for transportation applications," in *Proc. IEEE Applied Power Electronics Conf. and Exposition (APEC'16)*, 2016. doi: 10.1109/APEC.2016.7468355.

M. K. Zadeh, B. Zahedi, M. Molinas, and L. E. Norum, "Centralized stabilizer for marine DC microgrid," in *Proc. Industrial Electron. Conf. (IECON'13)*, 2013. doi: 10.1109/IECON.2013.6699667.

DieselNet, "IMO marine engine regulations," Oct. 2016. Accessed on: April 2, 2018. [Online]. Available: <https://www.dieselnet.com/standards/inter/imo.php>

L. W. Y. Chua, T. Tjahjowidodo, G. G. L. Seet, and R. Chan, "Implementation of optimization-based power management for all-electric hybrid vessels," *IEEE Access*, vol. 6, pp. 74,339–74,354, Nov. 2018. doi: 10.1109/ACCESS.2018.2883324.

A. J. Sorensen et al., "Toward safer, smarter, and greener ships: Using hybrid marine power plants," *IEEE Electrification Mag.*, vol. 5, no. 3, pp. 68–73, Sept. 2017. doi: 10.1109/MELE.2017.2718861.

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